

ASSIGNMENT 1

Q1.

SOLUTION:

```
kd3_98850_pranit> DESCRIBE S
-> ;
```

Field	Type	Null	Key	Default	Extra
S#	char(5)	NO		NULL	
Sname	char(20)	YES		NULL	
status	smallint	YES		NULL	
city	char(15)	YES		NULL	

4 rows in set (0.00 sec)

```
kd3_98850_pranit> SELECT * FROM S;
```

S#	Sname	status	city
S1	Smith	20	London
S2	Jones	10	Paris
S3	Blake	30	Paris
S4	Clark	20	London
S5	Adams	30	Athens

5 rows in set (0.00 sec)

Note : P denotes the Parts table.

CREATE TABLE P (P# CHAR(6) NOT NULL, Pname CHAR(20), color
CHAR(6), weight SMALLINT, city CHAR(15));

SOLUTION:

```
kd3_98850_pranit> DESCRIBE P;
```

Field	Type	Null	Key	Default	Extra
#P	char(6)	NO		NULL	
Pname	char(20)	YES		NULL	
color	char(6)	YES		NULL	
weight	smallint	YES		NULL	
city	char(15)	YES		NULL	

5 rows in set (0.00 sec)

ASSIGNMENT 1

```
kd3_98850_pranit> SELECT * FROM P;
+-----+-----+-----+-----+-----+
| #P | Pname | color | weight | city |
+-----+-----+-----+-----+-----+
| P1 | Nut   | Red   | 12      | London |
| P2 | Bolt  | Green | 17      | Paris  |
| P3 | Screw | Blue  | 17      | Rome   |
| P4 | Screw | Red   | 14      | London |
| P5 | Cam   | Blue  | 12      | Paris  |
| P6 | Cog   | Red   | 19      | London |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Note : J denotes the Jobs table.

CREATE TABLE J (J# CHAR(4) NOT NULL, Jname CHAR(10), City
CHAR(15));

SOLUTION:

```
kd3_98850_pranit> DESCRIBE J;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| J#    | char(4)   | NO   |     | NULL    |       |
| Jname | char(10)  | YES  |     | NULL    |       |
| City  | char(15)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

ASSIGNMENT 1

CREATE TABLE sp (S# CHAR(4) NOT NULL, P# CHAR(4) NOT NULL, J# CHAR(4) NOT NULL, QTY INT);
TABLE SP S# P# J# QTY =====

SOLUTION:

```
kd3_98850_pranit> DESCRIBE SP;
```

Field	Type	Null	Key	Default	Extra
S#	char(4)	NO		NULL	
P#	char(4)	NO		NULL	
J#	char(4)	NO		NULL	
QTY	int	YES		NULL	

4 rows in set (0.00 sec)

```
kd3_98850_pranit> SELECT * FROM SP;
```

S#	P#	J#	QTY
S1	P1	J1	200
S1	P1	J4	700
S2	P3	J1	400
S2	P3	J2	200
S2	P3	J3	200
S2	P3	J4	500
S2	P3	J5	600
S2	P3	J6	400
S2	P3	J7	800
S2	P5	J2	100
S3	P3	J1	200
S3	P4	J2	500
S4	P6	J3	300
S4	P6	J7	300
S5	P2	J2	200
S5	P2	J4	100
S5	P5	J5	500
S5	P5	J5	100
S5	P5	J7	100
S5	P6	J2	200
S5	P1	J4	100
S5	P3	J4	200
S5	P4	J4	800
S5	P5	J4	400
S5	P6	J4	500

25 rows in set (0.00 sec)

ASSIGNMENT 1

1. Create 'hr' database. Import hr-db.sql.
2. List all tables in hr database. Display contents of all tables (one by one).

```
root> SHOW TABLES;
+-----+
| Tables_in_hr |
+-----+
| countries     |
| departments   |
| employees     |
| job_history   |
| jobs          |
| locations     |
| regions       |
+-----+
7 rows in set (0.00 sec)
```

```
root> SELECT * FROM countries;
+-----+-----+-----+
| COUNTRY_ID | COUNTRY_NAME                                | REGION_ID |
+-----+-----+-----+
| AR         | Argentina                                  | 20        |
| AU         | Australia                                  | 40        |
| BE         | Belgium                                    | 10        |
| BR         | Brazil                                     | 20        |
| CA         | Canada                                     | 20        |
| CH         | Switzerland                               | 10        |
| CN         | China                                      | 30        |
| DE         | Germany                                    | 10        |
| DK         | Denmark                                    | 10        |
| EG         | Egypt                                      | 50        |
| FR         | France                                    | 10        |
| GB         | United Kingdom of Great Britain and Nort | 10        |
| IL         | Israel                                    | 30        |
| IN         | India                                      | 30        |
| IT         | Italy                                      | 10        |
| JP         | Japan                                      | 30        |
| KW         | Kuwait                                    | 30        |
| ML         | Malaysia                                   | 30        |
| MX         | Mexico                                    | 20        |
| NG         | Nigeria                                    | 50        |
| NL         | Netherlands                               | 10        |
| SG         | Singapore                                  | 30        |
| US         | United States of America                  | 20        |
| ZM         | Zambia                                    | 50        |
| ZW         | Zimbabwe                                  | 50        |
+-----+-----+-----+
25 rows in set (0.00 sec)
```

ASSIGNMENT 1

```
root> SELECT * FROM departments;
```

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
30	Purchasing	114	1700
40	Human Resources	203	2400
50	Shipping	121	1500
60	IT	103	1400
70	Public Relations	204	2700
80	Sales	145	2500
90	Executive	100	1700
100	Finance	108	1700
110	Accounting	205	1700
120	Treasury	NULL	1700
130	Corporate Tax	NULL	1700
140	Control And Credit	NULL	1700
150	Shareholder Services	NULL	1700
160	Benefits	NULL	1700
170	Manufacturing	NULL	1700
180	Construction	NULL	1700
190	Contracting	NULL	1700
200	Operations	NULL	1700
210	IT Support	NULL	1700
220	NOC	NULL	1700
230	IT Helpdesk	NULL	1700
240	Government Sales	NULL	1700
250	Retail Sales	NULL	1700
260	Recruiting	NULL	1700
270	Payroll	NULL	1700

27 rows in set (0.00 sec)

```
root> SELECT * FROM employees;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID			
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_
24000.00		NULL	NULL	90		PRES
101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_
17000.00		NULL	100	90		VP
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_
17000.00		NULL	100	90		VP
103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_
9000.00		NULL	102	60		PROG
104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_
6000.00		NULL	103	60		PROG
105	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_
4800.00		NULL	103	60		PROG
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_
4800.00		NULL	103	60		PROG
107	Diana	Nguyen	DNGUYEN	1.590.555.0107	2017-02-07	IT_
4200.00		NULL	103	60		PROG
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_
12008.00		NULL	101	100		MGR
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_
9000.00		NULL	108	100		ACCOUNT
110	John	Chen	JCHEN	1.515.555.0110	2015-09-28	FI_
8200.00		NULL	108	100		ACCOUNT
111	Ismael	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_
7700.00		NULL	108	100		ACCOUNT
112	Jose Manuel	Urman	JMURMAN	1.515.555.0112	2016-03-07	FI_
7800.00		NULL	108	100		ACCOUNT
113	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_
6900.00		NULL	108	100		ACCOUNT
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_
11000.00		NULL	100	30		MAN
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_
3100.00		NULL	114	30		CLERK

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```
root> SELECT * FROM job_history;
```

EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
101	2007-09-21	2011-10-27	AC_ACCOUNT	110
101	2011-10-28	2015-03-15	AC_MGR	110
102	2011-01-13	2016-07-24	IT_PROG	60
114	2016-03-24	2017-12-31	ST_CLERK	50
122	2017-01-01	2017-12-31	ST_CLERK	50
176	2016-03-24	2016-12-31	SA_REP	80
176	2017-01-01	2017-12-31	SA_MAN	80
200	2005-09-17	2011-06-17	AD_ASST	90
200	2012-07-01	2016-12-31	AC_ACCOUNT	90
201	2014-02-17	2017-12-19	MK_REP	20

```
10 rows in set (0.00 sec)
```

```
root> SELECT * FROM jobs;
```

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
AC_ACCOUNT	Public Accountant	4200	9000
AC_MGR	Accounting Manager	8200	16000
AD_ASST	Administration Assistant	3000	6000
AD PRES	President	20080	40000
AD_VP	Administration Vice President	15000	30000
FI_ACCOUNT	Accountant	4200	9000
FI_MGR	Finance Manager	8200	16000
HR_REP	Human Resources Representative	4000	9000
IT_PROG	Programmer	4000	10000
MK_MAN	Marketing Manager	9000	15000
MK_REP	Marketing Representative	4000	9000
PR_REP	Public Relations Representative	4500	10500
PU_CLERK	Purchasing Clerk	2500	5500
PU_MAN	Purchasing Manager	8000	15000
SA_MAN	Sales Manager	10000	20080
SA_REP	Sales Representative	6000	12008
SH_CLERK	Shipping Clerk	2500	5500
ST_CLERK	Stock Clerk	2008	5000
ST_MAN	Stock Manager	5500	8500

```
19 rows in set (0.00 sec)
```

ASSIGNMENT 1

```
root> SELECT * FROM locations;
+-----+-----+-----+-----+
| LOCATION_ID | STREET_ADDRESS | POSTAL_CODE | CITY |
| STATE_PROVINCE | COUNTRY_ID |
+-----+-----+-----+-----+
| 1000 | 1297 Via Cola di Rie | 00989 | Roma |
| NULL | IT |
| 1100 | 93091 Calle della Testa | 10934 | Venice |
| NULL | IT |
| 1200 | 2017 Shinjuku-ku | 1689 | Tokyo |
| Tokyo Prefecture | JP |
| 1300 | 9450 Kamiya-cho | 6823 | Hiroshima |
| NULL | JP |
| 1400 | 2014 Jabberwocky Rd | 26192 | Southlake |
| Texas | US |
| 1500 | 2011 Interiors Blvd | 99236 | South San Francisco |
| California | US |
| 1600 | 2007 Zagora St | 50090 | South Brunswick |
| New Jersey | US |
| 1700 | 2004 Charade Rd | 98199 | Seattle |
| Washington | US |
| 1800 | 147 Spadina Ave | M5V 2L7 | Toronto |
| Ontario | CA |
| 1900 | 6092 Boxwood St | YSW 9T2 | Whitehorse |
| Yukon | CA |
| 2000 | 40-5-12 Laogianggen | 190518 | Beijing |
| NULL | CN |
| 2100 | 1298 Vileparle (E) | 490231 | Bombay |
| Maharashtra | IN |
| 2200 | 12-98 Victoria Street | 2901 | Sydney |
| New South Wales | AU |
| 2300 | 198 Clementi North | 540198 | Singapore |
| NULL | SG |
| 2400 | 8204 Arthur St | NULL | London |
| NULL | GB |
| 2500 | Magdalen Centre, The Oxford Science Park | OX9 9ZB | Oxford |
| Oxford | GB |
| 2600 | 9702 Chester Road | 09629850293 | Stretford |
```

```
root> SELECT * FROM regions;
+-----+-----+
| REGION_ID | REGION_NAME |
+-----+-----+
| 50 | Africa |
| 20 | Americas |
| 30 | Asia |
| 10 | Europe |
| 40 | Oceania |
+-----+-----+
5 rows in set (0.00 sec)
```

ASSIGNMENT 1

2. Create 'sales' database. Import sales-db.sql.

```
root> SHOW TABLES;
+-----+
| Tables_in_sales |
+-----+
| customers       |
| orders          |
| salespeople     |
+-----+
3 rows in set (0.00 sec)
```

```
root> SELECT * FROM customers;
+-----+-----+-----+-----+-----+
| cnum | cname   | city    | rating | snum |
+-----+-----+-----+-----+-----+
| 2001 | Hoffman | London  | 100    | 1001 |
| 2002 | Giovanni | Rome    | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin  | 300    | 1002 |
| 2006 | Clemens  | London  | 100    | 1001 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
| 2007 | Pereira  | Rome    | 100    | 1004 |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

```
root> SELECT * FROM orders;
+-----+-----+-----+-----+-----+
| onum | amt     | odate      | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69   | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19  | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75   | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95  | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```


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```
root> SELECT * FROM salespeople;
+-----+-----+-----+-----+
| snum | sname | city | comm |
+-----+-----+-----+-----+
| 1001 | Peel | London | 0.12 |
| 1002 | Serres | San Jose | 0.13 |
| 1004 | Motika | London | 0.11 |
| 1007 | Rifkin | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

1. **How many orders are present in the database?**
10 orders
2. **How many customers are present in the database?**
7 customers
3. **How many salespeople are present in the database?**
5 salespeople.
4. **On which date order with highest amount is placed?**
The highest order amount is 9891.88, and it was placed on 1990-10-04.
5. **For which order salesman got maximum commission?**
The maximum commission is earned on order 3011.
6. **Which salesmen are handling more than two customers?**
No salesman is handling more than two customers..
7. **Which city has a single customer?**
Berlin.
8. **Which city has multiple salespeople?**
London has multiple salespeople.
9. **What is the name of customer and salesman of the maximum amount order?**
The maximum amount order is 3011. Customer: Clemens Salesman: Peel Amount: 9891.88
10. **Study relationship between the tables in sales database.**
Customers Orders: One customer can place many orders. Salespeople Orders: One salesperson can handle many orders. Salespeople Customers: One salesperson can handle many customers.